Classwork 09//Markov Decision Process, Remedial for Quiz 3

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Due Date:** April 14, 2026

1 Markov Decision Process (20 Points)

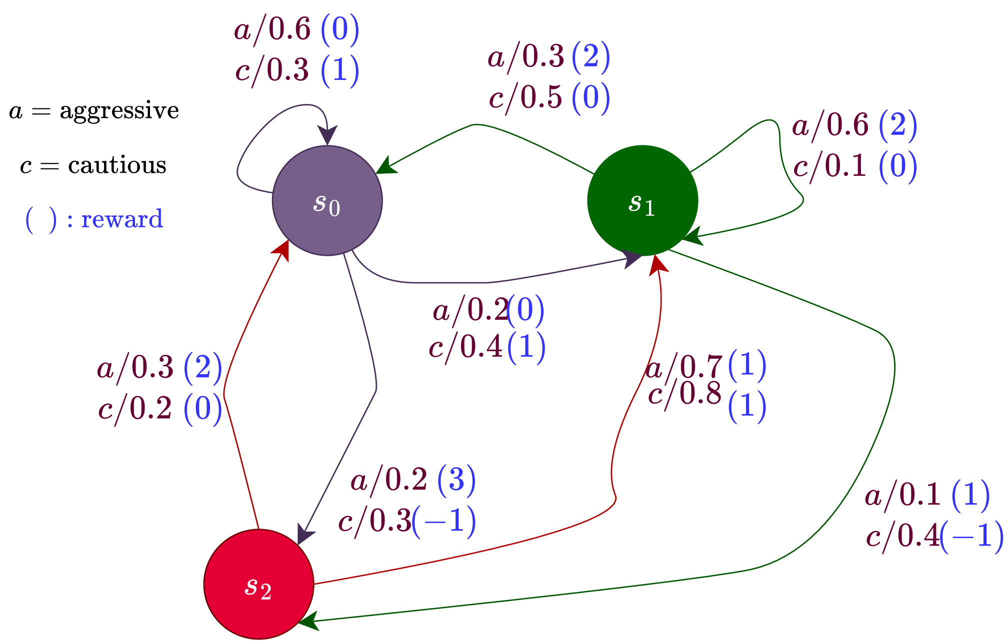

Figure 1: Q1

1. Expected immediate reward is given as

$$\bar{R}(s,a) = \sum_{s'} P(s'|s,a)R(s,a,s')$$

Calculate $\bar{R}(s_0,a)$ and $\bar{R}(s_0,c)$. Show your work. **(5 Points)**

2. Based solely on expected immediate reward, which action (aggressive or cautious) is preferred in each state? Does the action stay consistent across all three states? **(5 Points)**

3. Suppose the agent follows the always aggresive policy π_a starting from s_1. What is the expected reward after exactly one step? Repeat the same for always-cautious policy π_c.**(5 Points)**

4. Complete the transition table based on the diagram: **(5 Points)**

From	**Action**	**To**	**Probability**	**Reward**
s_0	a	s_0	0.6	0
s_0	a	s_1	0.2	0
s_0	a		0.2	3
s_0	c	s_0	0.3	
s_0	c		0.4	1
s_0		s_2	0.3	-1
s_1	a	s_0	0.3	2
s_1	a		0.6	2
s_1	a	s_2	0.1	1
s_1		s_0	0.5	0
s_1	c	s_1	0.1	0
s_1		s_2	0.4	-1
s_2	a	s_0	0.3	
s_2	a		0.7	1
s_2	c	s_0	0.2	0
s_2	c	s_1		1

Table 1: Full transition table. a = aggressive, c = cautious. Each row encodes $P(s' \mid s, a)$ and $R(s, a, s')$.

1.1 Solution

1.

$$\begin{aligned}
\bar{R}(s_0, a) &= \sum_{s'} P(s'|s_0, a) R(s_0, a, s') \\
& P(s_0|s_0, a) R(s_0, a, s_0) + P(s_1|s_0, a) R(s_0, a, s_1) + P(s_2|s_0, a) R(s_0, a, s_2) \\
&= 0.6 \times 0 + 0.2 \times 0 + 0.2 \times 3 = 0.6
\end{aligned}$$

$$\bar{R}(s_0, c) = \sum_{s'} P(s'|s_0, c)R(s_0, c, s')$$
$$P(s_0|s_0, c)R(s_0, c, s_0) + P(s_1|s_0, c)R(s_0, c, s_1) + P(s_2|s_0, c)R(s_0, c, s_2)$$
$$= 0.3 \times 1 + 0.4 \times 1 + 0.3 \times (-1) = 0.3 + 0.4 - 0.3 = 0.4$$

2. For state s_0, $\bar{R}(s_0, a) > \bar{R}(s_0, c)$, so aggresive action is preferred.

$$\bar{R}(s_1, c) = \sum_{s'} P(s'|s_1, c)R(s_1, c, s')$$
$$P(s_0|s_1, c)R(s_1, c, s_0) + P(s_1|s_1, c)R(s_1, c, s_1) + P(s_2|s_1, c)R(s_1, c, s_2)$$
$$= 0.5 \times 0 + 0.1 \times 0 + 0.4 \times (-1) = -0.4$$

$$\bar{R}(s_1, a) = \sum_{s'} P(s'|s_1, a)R(s_1, a, s')$$
$$P(s_0|s_1, a)R(s_1, a, s_0) + P(s_1|s_1, a)R(s_1, a, s_1) + P(s_2|s_1, a)R(s_1, a, s_2)$$
$$= 0.3 \times 2 + 0.6 \times 2 + 0.1 \times 1 = 0.6 + 1.2 + 0.1 = 1.9$$

For state s_1, $\bar{R}(s_1, a) > \bar{R}(s_1, c)$, so aggresive action is preferred.

$$\bar{R}(s_2, c) = \sum_{s'} P(s'|s_2, c)R(s_2, c, s')$$
$$P(s_0|s_2, c)R(s_2, c, s_0) + P(s_1|s_2, c)R(s_2, c, s_1) + P(s_2|s_2, c)R(s_2, c, s_2)$$
$$= 0.2 \times 0 + 0.8 \times 1 = 0.8$$

$$\bar{R}(s_2, a) = \sum_{s'} P(s'|s_2, a)R(s_2, a, s')$$
$$P(s_0|s_2, a)R(s_2, a, s_0) + P(s_1|s_2, a)R(s_2, a, s_1) + P(s_2|s_2, a)R(s_2, a, s_2)$$
$$= 0.3 \times 2 + 0.7 \times 1 = 0.6 + 0.7 = 1.3$$

For state s_2, $\bar{R}(s_2, a) > \bar{R}(s_2, c)$, so aggresive action is preferred.

3. For always aggressive,starting from s_1, the one-step expected reward equals $\bar{R}(s_1, a)$ directly:

$$\mathbb{E}[R|s_1, \Pi_a] = 0.6 \times 2 + 0.3 \times 2 + 0.1 \times 1 = 1.2 + 0.6 + 0.1 = 1.9$$

For always cautious,starting from s_1, the one-step expected reward equals $\bar{R}(s_1, c)$ directly:

$$\mathbb{E}[R|s_1, \Pi_c] = 0.1 \times 0 + 0.4 \times (-1) + 0.5 \times 0 = -0.4$$

From	**Action**	**To**	**Probability**	**Reward**
s_0	a	s_0	0.6	0
s_0	a	s_1	0.2	0
s_0	a	$\boldsymbol{s_2}$	0.2	3
s_0	c	s_0	0.3	**1**
s_0	c	$\boldsymbol{s_1}$	0.4	1
s_0	$\boldsymbol{c}$	s_2	0.3	-1
s_1	a	s_0	0.3	2
s_1	a	$\boldsymbol{s_1}$	0.6	2
s_1	a	s_2	0.1	1
s_1	$\boldsymbol{c}$	s_0	0.5	0
s_1	c	s_1	0.1	0
s_1	$\boldsymbol{c}$	s_2	0.4	-1
s_2	a	s_0	0.3	**2**
s_2	a	$\boldsymbol{s_1}$	0.7	1
s_2	c	s_0	0.2	0
s_2	c	s_1	**0.8**	1

Table 2: Full transition table. a = aggressive, c = cautious. Each row encodes $P(s' \mid s, a)$ and $R(s, a, s')$.

4.

2 Quiz 3 Remedial

Please go through the solution of Quiz 3 posted on Canvas, and rework on the solution to every problem, and provide justification for the each of the solution. **(10 Points)**